Samanvay Malapally Sudhakara

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Quantitative Researcher with **2+ years** of experience leveraging advanced analytics & machine learning to solve complex challenges in finance, technology & business operations. CFA Level 3 Candidate. Adept at deriving actionable insights from complex datasets using statistical analysis & predictive modeling, & at communicating them to stakeholders. Proficient in cloud technologies & experienced in deploying scalable data & LLM-inference solutions. Strong quantitative & mathematical skills, including statistical modeling, stochastic calculus, numerical methods, Dynamic Programming, Game Theory & expertise in building algorithmic trading systems & developing high-frequency trading (HFT) strategies.

TECHNICAL SKILLS

Languages	Python, Rust, SQL, C++, R, Linux Shell, Git
Data Visualisation	Numpy, Pandas, Seaborn, Matplotlib, Plotly, ggplot2, Tableau, Power BI, Streamlit, Excel
Machine Learning	Regression, Classification, Decision Trees, Ensemble Trees, KNN, K-Fold CV, PCA, LDA, SVM, ANN, CNN, RNN, LSTM, GRU, Transformers, DQN, DDPG, VAE, GANs, U-NET Diffusion Models, NLP, LLM/SLM pretraining, finetuning (RLHF)
ML Packages	Pytorch, Tensorflow, Keras, Scikit-learn, OpenCV, Transformers, Huggingface-hub, NLTK, Word2Vec, spaCy, LangChain
LLM & Inference	vLLM, NVFP4/FP8 Quantization, Speculative Decoding (DFlash), MCP, Pydantic AI, LlamaIndex, Qdrant, Weaviate, RAG/GraphRAG, Phoenix/OpenInference, Cloudflare Tunnel, Tailscale, CUDA/ARM64, Claude Code, Codex
Cloud & Databases	AWS (S3, EMR, EC2, Redshift, Sagemaker), GCP (Compute Engine, GCS, Vertex AI, Cloud Run, BigQuery)
Big Data, CI/CD	MySQL, Apache (Hadoop, Airflow, Spark, Kafka), MapReduce, Gitlab, Github, Kubernetes, Docker, SnowFlake
Quantitative Skills	Quantitative Finance, Stochastic Calculus, Derivatives, Hypothesis Testing, Statistical Analysis, Risk Management, Market Microstructure, Algorithmic Trading, Interest Rate Models, Game theory, Dynamic Programming & Reinforcement Learning

PROFESSIONAL EXPERIENCE

Quantitative Associate, *Northwestern Mutual Investment Management Company LLC, WI* — Oct 2024 - Current

- Led the department's shift to a Python/Streamlit stack for dashboards & tooling across the full fixed income spectrum: IG/HY Corporates, EM sovereign & corporate credit, MBS, CMBS, CLOs, ABS, Munis, Rates & Macro & fixed income derivatives on a $140B+ AUM book.
- Built tooling that consolidates portfolio analytics from BlackRock Aladdin, Bloomberg & internal sources into a unified Snowflake database modeled with dbt, establishing a single source of truth for the analytics layer.
- Architected a multi-app Streamlit interface mimicking Tableau's UI, fronted by a semantic layer that lets PMs aggregate & slice the portfolio by any field & render it as a table, matrix, time series, scatter or treemap with fully customizable axes and implemented bottom-up performance attribution computed from run-time aggregations in seconds, driven by the same semantic-layer queries underlying the exploration tool.
- Developed trader tools that parse Markit & Neptune quotes into an actionable format & link directly to the portfolio analytics, collapsing the ideation-to-trade workflow from hours to minutes.
- Automated end-of-day report ingestion to deliver data 4 hours earlier each day, & built parallel Streamlit tooling to compare & analyze Agency MBS pools & cohorts from eMBS loan-level data.

PROJECTS

High Frequency Trading Backtesting Pipeline with RL Trading Bot

- Constructed Data Parsers for tick, depth update & market by order data from sources such CME Globex DataSet, IEX , Binance & Oanda along with data cleaning & exploratory data analysis to identify outliers, missing data & visualise the data (link)
- Simplified running strategy backtests by building backtesting pipelines for running on command line with Linux shell scripts that seamlessly integrate various C++, python scripts necessary to run backtests, process results & tune hyperparameters (link)
- Engineered DQN & DDPG models in C++ for a market-taking agent & a market-making agent with PyTorch C++ API for the model & Sqlite for experience replay store & backtested using previously crafted pipelines (link)

Agentic Equity Research Pipeline on Self-Hosted LLM Inference

- Deployed a 35B-parameter MoE model (Qwen3.5-A3B) on an NVIDIA DGX Spark (GB10/ARM64, 128GB unified memory) via vLLM with NVFP4 quantization & DFlash speculative decoding, fronted by a Pydantic AI + MCP agent gateway (streamable HTTP) exposed via Cloudflare Tunnel with Access JWT for zero-open-port ingress & instrumented with Phoenix/OpenInference
- Engineered an event-driven BSE/NSE filing sentinel using a small LLM for materiality classification that triggers per-ticker research updates, with Firestore hash-based deduplication & Pub/Sub handoff, feeding a LlamaIndex + Qdrant + FastEmbed RAG system over a DuckDB/Parquet data lake on GCS

CERTIFICATIONS

- **Bloomberg Market Concepts** *Bloomberg For Education*
- **Machine Learning Specialisation** *DeepLearning.AI*
- **Deep Learning Specialisation** *DeepLearning.AI*

EDUCATION

MS Financial Engineering (GPA 3.77/4), *University of Illinois at Urbana-Champaign* — Champaign, Illinois — May 2024

BE Mechanical Engineering, *Ramaiah Institute of Technology* — Bangalore, India — July 2021